Name: ______________________________

Exit quiz

Perimeter of composite shapes

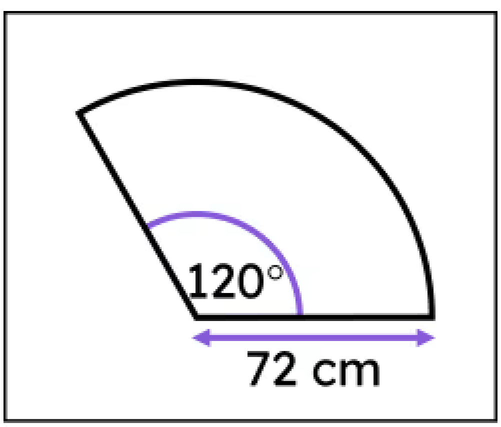
1 A quarter-circular sector is cut out of a circle. The original circle had a perimeter of 60π cm. What is the arc length of this sector? Tick **1** correct answer

- [ ] 15π cm
- [ ] $15\pi + 30$ cm
- [ ] $15\pi + 60$ cm

2 A semicircular sector is cut out of a circle. The radius of the original circle is 31 cm, and its circumference is 194.8 cm (rounded to 1 decimal place). What is the perimeter of the sector? Tick **2** correct answers

- [ ] 31π
- [ ] 97.4 cm
- [ ] 97.4 + 31 cm
- [ ] 97.4 + 62 cm
- [ ] 159.4 cm

3 Calculate the perimeter of this sector, giving your answer in cm and rounding your answer to 1 decimal place. Fill in the blank



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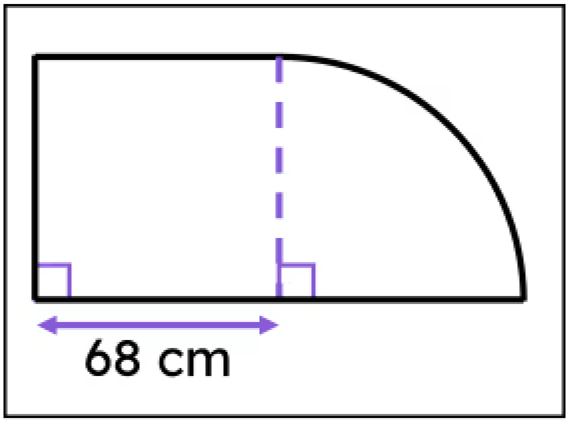
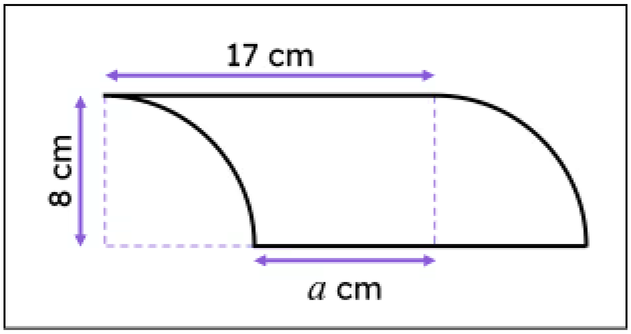
4 This composite shape is composed of a circular sector and a square. Which of these statements are true about the perimeter of this composite shape that has been broken into component lengths. Tick **2** correct answers



- [ ] There are three line segments of 68 cm.
- [ ] There are four line segments of 68 cm.
- [ ] There are five line segments of 68 cm.
- [ ] The length of the arc is 68π cm.
- [ ] The length of the arc is 34π cm.

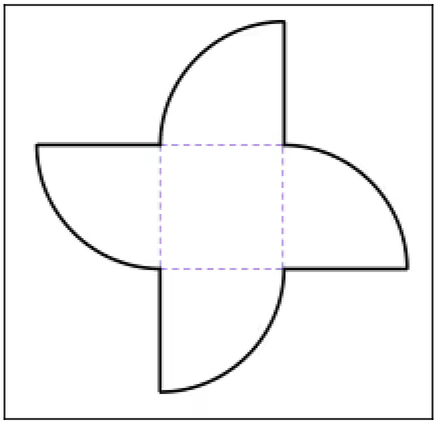
5 This composite shape is composed of two quarter-circles and a rectangle. Match the statement about the component parts of the composite shape's perimeter to the correct value. Write the correct letter in each box



a	value of a
b	radius of each quarter-circle
c	total length of line segments
d	total length both arcs
e	total perimeter of shape

	34 cm
	8 cm
	59.1 cm
	9 cm
	8π cm

6 The square at the centre of this composite shape has an area of 441 cm². Find the perimeter of the composite shape in cm, rounded to the nearest integer. Fill in the blank



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